

Negation and contrapositive

Guided notes and practice

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The question to ask: what would make the statement false? A negation is true exactly when the original statement is false. Keep the same domain (real numbers, integers, students, and so on). Write the negation before deciding whether either statement is actually true.

1. Start with a single condition

For a real number x , the statement $x > 3$ fails when $x \leq 3$. Equality must be included: $x = 3$ does not satisfy $x > 3$. Similarly,

$$\neg(x \geq 3) \iff x < 3, \quad \neg(x \neq 3) \iff x = 3.$$

2. Make “and” or “or” fail

If both parts are required, one failure is enough. If either part is allowed, both must fail:

$$\neg(P \wedge Q) \iff \neg P \vee \neg Q, \quad \neg(P \vee Q) \iff \neg P \wedge \neg Q.$$

For an integer n , “ n is positive and even” is false when $n \leq 0$ or n is odd. Ask yourself which part you could break to make the whole sentence fail.

3. Turn “every” into “some”

One exception disproves a claim about every member of a set. A claim that *some* member works fails only if none works:

$$\neg(\forall x \in D, P(x)) \iff \exists x \in D \text{ such that } \neg P(x),$$

$$\neg(\exists x \in D \text{ such that } P(x)) \iff \forall x \in D, \neg P(x).$$

Example: “Every integer n has $n^2 \geq 0$ ” negates to “There is an integer n with $n^2 < 0$.” That negation is false, but it is still the correct negation. With two quantifiers, change each in its original order: $\neg(\forall x \exists y R(x, y))$ becomes $\exists x \forall y \neg R(x, y)$.

4. An implication has two different relatives

“If P , then Q ” fails only when P happens but Q does not. Its *negation* is $P \wedge \neg Q$. Its *contrapositive* is another if-then statement, $\neg Q \Rightarrow \neg P$, with the same truth value as the original:

$$\neg(P \Rightarrow Q) \iff P \wedge \neg Q, \quad (P \Rightarrow Q) \iff (\neg Q \Rightarrow \neg P).$$

For every integer n : “If $4 \mid n$, then n is even.” Its negation is “There is an integer divisible by 4 that is odd.” Its contrapositive is “If n is odd, then $4 \nmid n$.” The converse, “If n is even, then $4 \mid n$,” is different and fails at $n = 2$.

Before you answer: name the domain, identify the condition that would fail, and check whether you need a negation (a way the statement fails) or a contrapositive (an equivalent if-then statement). “ P only if Q ” means $P \Rightarrow Q$.

Practice questions

First write the logical form, then check it with a simple example where useful.

A. Basic negations

Question 1

For a real number x , negate each statement: (a) $x > 7$; (b) $x \neq 2$.

Solution. (a) $x \leq 7$. The boundary $x = 7$ makes $x > 7$ false. (b) $x = 2$, the only way for $x \neq 2$ to fail.

Question 2

For an integer n , negate: “ n is positive and even.”

Solution. $n \leq 0$ or n is odd. The original requires both conditions, so breaking either one is enough.

Question 3

For real numbers a, b , negate: “ $a < 1$ or $b \geq 4$.”

Solution. $a \geq 1$ and $b < 4$. An “or” statement fails only when both alternatives fail.

Question 4

For an integer n , negate: “ n is divisible by 3 but not by 5.”

Solution. $3 \nmid n$ or $5 \mid n$. Here “but” joins two conditions in the same way as “and”.

B. Every, some and quantifiers

Question 5

Negate: “Every integer $n \geq 1$ satisfies $n^2 > n$.” If the negation is true, give a value of n that shows it.

Solution. There exists an integer $n \geq 1$ with $n^2 \leq n$. The witness $n = 1$ works because $1^2 = 1$. Hence the original statement is false.

Question 6

Negate: “There is a real number x such that $x^2 + 1 = 0$.” Decide which statement is true.

Solution. For every real x , $x^2 + 1 \neq 0$. This negation is true: $x^2 \geq 0$, so $x^2 + 1 \geq 1$ for every real x .

Question 7

Negate: “For every integer n , there is an integer $m > n$.” Decide which statement is true.

Solution. There exists an integer n such that every integer m satisfies $m \leq n$. This negation is false: whatever n is, $m = n + 1$ is an integer greater than n . Thus the original claim is true.

Question 8

Negate: “Some student solved every question on the worksheet.”

Solution. Every student has at least one question on the worksheet that they did not solve. In symbols, $\neg(\exists s \forall q S(s, q))$ is $\forall s \exists q \neg S(s, q)$, where $S(s, q)$ means student s solved question q .

C. Implications and contrapositives

Question 9

For every integer n : “If $8 \mid n$, then n is even.” Write (a) the negation and (b) the contrapositive.

Solution. (a) There exists an integer n divisible by 8 that is odd. This is false. (b) For every integer n , if n is odd, then $8 \nmid n$. This is true, just like the original, since $8k = 2(4k)$ is even.

Question 10

For every real x : “If $x^2 > 9$, then $x > 3$.” (a) Find a counterexample. (b) Negate the statement. (c) Write its contrapositive.

Solution. (a) $x = -4$ has $x^2 = 16 > 9$ but $x \leq 3$. (b) There exists a real x such that $x^2 > 9$ and $x \leq 3$. (c) For every real x , if $x \leq 3$, then $x^2 \leq 9$. The contrapositive is also false at $x = -4$, as equivalent statements must be.

Question 11

“An integer is divisible by 12 only if it is divisible by 3.” (a) Rewrite this as an if-then statement. (b) Give its contrapositive. (c) Is its converse true?

Solution. (a) For every integer n , if $12 \mid n$ then $3 \mid n$. (b) For every integer n , if $3 \nmid n$ then $12 \nmid n$. (c) The converse, “if $3 \mid n$ then $12 \mid n$,” is false: take $n = 3$.

D. Short proofs

Question 12

For an integer n , prove by contrapositive: “If n^2 is odd, then n is odd.”

Solution. The contrapositive says: if n is even, then n^2 is even. Write $n = 2k$ for some integer k . Then $n^2 = 4k^2 = 2(2k^2)$, which is even. Therefore the contrapositive, and hence the original statement, is true.

Question 13

For integers a, b , prove by contrapositive: “If ab is odd, then both a and b are odd.”

Solution. The contrapositive is: if a is even or b is even, then ab is even. If $a = 2k$, then $ab = 2(kb)$; if $b = 2k$, then $ab = 2(ka)$. In either case ab is even. This proves the contrapositive and therefore the original implication.

Final check: Did you keep the domain? Did equality switch to the correct side? Did each “every” become “some” (and each “some” become “every”)? For an implication, have you kept its negation separate from its contrapositive?